Prime separation for Diophantine ZFC consistency

A polynomial in thirteen unrestricted integers
of degree 1,392, evaluated in 205 binary operations

Prepared for Vladimir Reshetnikov and the ProveIt repository
AI-assisted mathematical research, with exact arithmetic artifacts

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Abstract

We construct an explicit ordinary polynomial in thirteen unrestricted integer unknowns whose lack of integer zeros is provably equivalent in ZFC to syntactic consistency of ZFC, by composition with the identified native-compiler and refutation-calibration theorems in ProveIt. Its exact total degree is $1,392$, and the supplied shared arithmetic circuit uses 205 binary additions, subtractions, and multiplications. All six fixed integer coefficients are supplied as exact byte strings.

The new arithmetic method separates several algebraic roots by their orders at the fixed prime 3. A rational sum then forces each root to be rational, and monicity or an elementary Pell descent recovers integrality. This permits four Pell and quotient witnesses to share one integer coordinate. A final step incorporates a root of the positivity guard itself; a syntactic oddness condition establishes the necessary nondegeneracy before that guard is recovered. Companion equations have fourteen, fifteen, and sixteen unrestricted integer unknowns at exact degrees $696,244,108$, in $151,137,132$ operations respectively.

The work retains the source-34 native computational encoding. It does not claim a globally smallest equation, literature-wide priority, consistency of ZFC, or a new complete proof-assistant certificate. The new local reductions, their canonical completeness arguments, their exact degrees, and their literal polynomial implementations are given in full.

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1 Results and the meaning of the comparison

The requested object is a polynomial with fixed integer coefficients and a specified domain of unknowns. We use syntactic consistency:

$$\mathrm{Con}(\mathsf{ZFC}) \quad \equiv \quad \neg\exists p \in \mathbb{N}\ \mathrm{Ref}_{\mathsf{ZFC}}(p),$$

where $\mathrm{Ref}_{\mathsf{ZFC}}$ is the fixed recursive predicate recognizing finite ZFC refutations used by the repository's calibrated endpoint. A theorem that ZFC proves an equivalence between this sentence and unsolvability is different from a proof of either sentence separately.

Theorem 1.1 (Main construction)**.** *For the six fixed integers specified in Section 2.2, the explicit polynomial P_{13} of Section 2 satisfies*

$$\mathsf{ZFC} \vdash \left(\neg\exists \mathbf{x} \in \mathbb{Z}^{13}\ P_{13}(\mathbf{x}) = 0\right) \iff \mathrm{Con}(\mathsf{ZFC}).$$

Its exact total degree is $1,392$*. The delivered arithmetic circuit evaluates it in* 205 *binary operations. These assertions use the inherited source contract stated in Section 3.*

Three companion polynomials retain different numbers of coordinates. They address different degree and evaluation-cost preferences; the thirteen-variable equation does not dominate them in every measure.

Polynomial	Integer unknowns	Exact degree	Operations	Final form
P_{13}	**13**	**1,392**	**205**	One norm
P_{14}	14	696	151	Norm and guard
P_{15}	15	244	137	Norm and two rows
P_{16}	16	108	132	Four squared rows

Table 1: All unknowns range over unrestricted integers. Operations count every live binary $+, -, \times$, including squaring and multiplication by fixed numerals. Shared reuse and fixed integer leaves are free.

The comparison is against the inspected repository snapshot `cfbf118b4ad9b2c651fe96b571d9415261766328`, including relevant incoming archives, rather than only an earlier collection README. In the delivered collection, source 67 gives a thirteen-integer equation of degree $10,461,744$ in 519 operations; a different earlier thirteen-integer point uses 518 operations at a larger degree [4, 1]. The new 13/1392/205 point substantially improves that particular domain and variable-count slice. This is a bounded comparison to supplied constructions, not a lower-bound theorem about all polynomials.

The repository also contains equations in ten unrestricted integers and nine positive integers. Those remain distinct points in the comparison: fewer variables, the sign restriction on their domains, their degrees, and their evaluation lengths must all be recorded. We additionally supply small exact circuit improvements to current ten-integer equations in Section 13; they preserve the existing polynomials.

1.1 What changes mathematically

Previous root mergers needed variable scales, large separating bounds, or carefully engineered congruences to prevent irrational roots from canceling. Here a fixed prime already present in the

source arithmetic distinguishes their orders. For four relevant roots the orders are

$$2, \quad 1, \quad 0, \quad -v_3(H).$$

A fifth guard root has order at least 3, or is zero. Every subset containing an old root therefore has a uniquely smallest order. A finite sign-product argument shows that such a subset cannot cancel. A Galois argument converts this fact into rationality of individual roots.

There are two additional requirements. Rationality must be strengthened to integrality, and eliminating a guard must not remove a premise needed to recover that guard. The proof handles the first by monic square equations and an adjacent-coefficient Pell descent. It handles the second by making c odd in the syntax of the polynomial and proving the necessary bounds before using any positivity guard.

1.2 What remains inherited

This is an arithmetic reduction of an established source encoding. The fixed coefficient tuple still represents the native language associated with the K215 blank computation. The new thirteen-variable equation does not invoke the Jones or Sun small-variable finalizer, but its underlying computational source has not been replaced by a direct encoding of set-theoretic proof syntax. The source contract below makes that dependency explicit. No finite numerical test is presented as a proof of an unbounded computation or of consistency.

2 The complete thirteen-variable polynomial

2.1 Exactly thirteen independent inputs

Let

$$\boxed{j, C, z, h, r, v, w, c_0, T, s, p_2, p_3, x \in \mathbb{Z}.}$$

Every other symbol in this section is a polynomial abbreviation, except for the single rational abbreviation B, whose denominator is cleared in the final formula. None is an additional quantified unknown.

Define

$$\begin{aligned}
&J = 6j + 1, && c = 2c_0 + 1, && q = \beta J + 1, \\
&W = 4(3w + 1), && X = qW, && Y = (2v + 1)q^3, \\
&E = XY, && a = (X + 1)Y, && H = 4a + 3, \\
&\Delta = a^2 + H, && L = (2r + 1)c^2, && F_2 = 1 + \Delta L^2, \\
&Q = \Delta^2 L^2, && \delta = Q - 1.
\end{aligned} \tag{1}$$

The native transport and index expressions are

$$F = (K_0 + W)C - z(q - 1), \tag{2}$$

$$\begin{aligned}
&R = \big(q(q - F) - C\big)(q^2 - 1) + (M_C + qM_F)J, \\
&k = R + 1 + hE, && \eta = c - Yk, && u = EYk, \\
&A = u(u + k) + 1, && V = F_2(cT - R) - c.
\end{aligned} \tag{3}$$

Set

$$D_0 = X + ac, \qquad D_m = 1 + \Delta c^2, \qquad f_0 = (a+2)L, \tag{4}$$
$$B = \frac{QV^2 - 1}{\delta}.$$

The determinant, interval product, and guard radicand are

$$d = (a_0F)(b_0C) - (q - a_0F - b_0C)^2, \tag{5}$$
$$T_\eta = \eta(k - \eta),$$
$$\Omega = dT_\eta - (s^2 - d - T_\eta + 1)^2 - 1 - p_2^2 - p_3(p_3 + 1). \tag{6}$$

Here a_0, b_0 are fixed coefficient scales, not unknowns. Finally put

$$Z = H\big[x + 18f_0 + 3(2u + k) + 2V\big] + 2D_0. \tag{7}$$

Definition 2.1 (Finite polynomial norm)**.** For $m \geq 1$, define

$$\mathcal{N}_m(Z; A_1, \ldots, A_m) = \prod_{\epsilon \in \{-1,1\}^m} \left(Z + \sum_{i=1}^{m} \epsilon_i \sqrt{A_i} \right).$$

This notation denotes the resulting polynomial in $\mathbb{Z}[Z, A_1, \ldots, A_m]$.

Changing the sign of any formal square root permutes the factors. Every exponent of that root is therefore even after multiplication. Consequently Definition 2.1 specifies an ordinary polynomial; it does not require choosing analytic square-root branches.

The promised equation is

$$\boxed{P_{13} = \delta^{16}\mathcal{N}_5(Z;\ 324H^2F_2,\ 36H^2A,\ 4H^2B,\ 4D_m,\ 2916H^2\Omega) = 0.} \tag{8}$$

The degree in any one radicand of $\mathcal{N}_5$ is at most 16. Thus δ^{16} clears every occurrence of the displayed denominator. An evaluator using only integer addition, subtraction, and multiplication is specified in Section 10. In particular, the delivered polynomial has no division operation, variable exponent, unspecified parameter, or implicit extra witness.

2.2 The fixed coefficients

The complete source-34 tuple is used, without mixing its ports with another native layout:

$$\beta = 2^{1064921} - 1, \qquad a_0 = 2^{1113}, \qquad b_0 = 2^{666104}.$$

The other three fixed numerals K_0, M_C, M_F are supplied literally in the archive. They are the authentic finite native-compiler outputs of the source-34 layout. M_F already contains the prescribed shift by β; applying that shift a second time would change the equation.

If a decompressed payload consists of bytes $b_0, \ldots, b_{n-1}$, its value is exactly $\sum_{\ell<n} b_\ell 256^\ell$. This convention, the literal payloads, the source pins, and the finite rebuilding recipe bind all six ports to fixed integers. The table measures the coefficient inputs to the circuit; it is not the size of a fully expanded polynomial.

Two elementary fixed facts used below are

$$\beta > 17, \qquad \beta \equiv 1 \pmod 3, \qquad \beta \text{ odd}.$$

They follow directly from the exponent 1064921, which is odd. No numeral is produced by running a refutation search.

Coefficient	Bits	Raw bytes	Role
β	1,064,921	133,116	Native base minus one
K_0	1,065,043	133,131	Fixed transport coefficient
M_C	1,064,921	133,116	Native low mask
M_F	1,064,922	133,116	Already shifted high mask
a_0	1,114	140	Determinant scale
b_0	666,105	83,264	Determinant scale
Total		615,883	10,607 bytes compressed

Table 2: Exact unsigned little-endian payloads, compressed individually with gzip. The binding ledger records full raw and compressed SHA-256 hashes.

3 The inherited source contract

The new proof has a small arithmetic interface to the existing work. Stating that interface is necessary: a local polynomial identity alone does not identify a zero with a ZFC refutation.

3.1 Soundness of the signed native interface

Sources 70 and 71 provide, on the same source-34 tuple, the residue-oriented signed arithmetic system with the calibrated determinant guard [2, 6, 7]. Its variables and quadratic registers can be projected to the compact interface used here. The retained conditions include the transport and index equations, first and main Pell norms, an ordinary strong norm, two auxiliary quotient congruences, the auxiliary norm, and the sign guards

$$d > 0, \qquad \eta(k-\eta) > 0.$$

Their theorem states that an unrestricted integer solution yields a finite syntactic ZFC refutation. The source proof restores its earlier square guard only after a signed Pell argument recovers the appropriate power of two; the present work does not reverse that logical order.

For the fifteen- and sixteen-variable companions we use the later seventeen-integer CRT parent from the incoming Pell-congruences archive [9]. That source proves its quotient compression before its low-degree projections. The ordinary signed Pell and rank proofs remain part of that parent's theorem.

3.2 Canonical completeness

A refutation supplies canonical source-34 witnesses with

$$\begin{gathered} q = 2^t, \quad t \text{ odd}, \quad R \geq 3, \quad R \equiv 3 \pmod 4, \\ X = 2^R, \quad Y \text{ an odd positive multiple of } q^3, \quad c = \psi_{a+2}(R) > 2, \\ \eta > 0, \quad k - \eta > 0, \quad d > 0. \end{gathered} \tag{9}$$

Here

$$\chi_b(n) + \psi_b(n)\sqrt{b^2-1} = (b + \sqrt{b^2-1})^n$$

defines the standard Pell polynomials for $b \geq 2$.

The source-71 determinant proof applies to the native F, C, q words throughout the canonical family, including its permitted rank tuning. The choice of later strong and auxiliary Pell witnesses

does not change those native words or the interval product $\eta(k-\eta)$. This freedom is what permits the normalized completeness construction in Section 8.

3.3 Scope of the logical claim

The existing endpoint connects the K215 blank run with the specified syntactic refutation predicate. The native compiler connects that run with the source arithmetic interface. This article proves the new unbounded implications between that interface and the literal new polynomials. All arguments use ordinary finite arithmetic, induction, polynomial algebra, and standard finite extensions of $\mathbb{Q}$, and can be formalized in ZFC.

The archive supplies exact finite implementations and mathematical proofs. It does not export an encoded derivation inside the repository's object-language ZFC calculus, or transfer the formal status of a different polynomial to these new ones. Internal independent reviews are identified as AI-assisted reviews rather than external peer review.

4 Separation of algebraic roots at the prime three

The useful principle concerns rational squares of algebraic numbers. Write $v_3(r)$ for the exponent of 3 in a nonzero rational number r, and write $\mathbb{Z}_{(3)}$ for rationals with denominator prime to 3.

Lemma 4.1 (No cancellation with a unique least order)**.** *Let $\alpha_1, \dots, \alpha_n$ be nonzero algebraic numbers with rational squares. Suppose every subset with at least two members has a unique member whose squared 3-adic order is smallest, and that smallest order is even. Then no nonempty signed sum of the α_i vanishes.*

Proof. A singleton does not vanish. In a larger subset let $2e$ be the unique smallest squared order, and divide each root by 3^e. Denote the normalized least root by β_0, and the others by $\beta_1, \dots, \beta_s$. Then β_0^2 is a unit of $\mathbb{Z}_{(3)}$, and each β_i^2, $i > 0$, belongs to $3\mathbb{Z}_{(3)}$.

Consider the finite product

$$U = \prod_{\epsilon \in \{-1,1\}^s} \left(\beta_0 + \sum_{i=1}^{s} \epsilon_i \beta_i \right).$$

Sign invariance in the β_i, and simultaneous negation when changing β_0, show that U is a polynomial with integer coefficients in all the squared roots. Reduction in $\mathbb{Z}_{(3)}/3\mathbb{Z}_{(3)}$ therefore gives

$$U \equiv (\beta_0^2)^{2^{s-1}} \not\equiv 0 \pmod 3.$$

Thus none of the factors can vanish. Every signed sum under consideration is, up to an overall sign, one of these factors. □

Corollary 4.2 (Rational extraction)**.** *Under the hypotheses of Lemma 4.1, if a signed sum of the roots is rational, each root is rational.*

Proof. The field generated by finitely many square roots of rational numbers is a finite Galois extension of $\mathbb{Q}$. Each automorphism changes some root signs. Subtracting its image of the rational sum gives twice a signed sum of exactly the changed nonzero roots. The lemma excludes such a nonempty sum. Hence every automorphism fixes every root, and the fixed field is $\mathbb{Q}$. □

We will use four squared orders

$$4, \quad 2, \quad 0, \quad -2v_3(H), \qquad v_3(H) \geq 1, \tag{10}$$

together with a fifth squared order at least 6. If the fifth root is zero, it is already rational and can be omitted. If it is nonzero, its order may be odd; this causes no problem because every subset containing an old root has a unique even minimum, and a subset consisting only of the fifth root is a singleton.

This proof uses orders of rational numbers and a finite sign product. It does not need a developed theory of valuations on algebraic extensions. Nor does it assume that the roots are real: negative radicands are allowed until rational extraction excludes them at a zero.

5 Integrality and the joint guard

Lemma 5.1 (Rational square integrality)**.** *If $r \in \mathbb{Q}$ and $r^2 \in \mathbb{Z}$, then $r \in \mathbb{Z}$.*

Proof. Write $r = a/b$ in lowest terms, $b > 0$. The divisibility $b^2 \mid a^2$ forces $b = 1$. □

The auxiliary Pell root needs a different argument because its square is initially a rational number.

Lemma 5.2 (Adjacent-coefficient descent)**.** *Let $Q > 1$ and V be integers. If $y \in \mathbb{Q}$ satisfies*

$$QV^2 - (Q-1)y^2 = 1,$$

then $y \in \mathbb{Z}$.

Proof. Write $y = a/m$ in lowest terms with $m > 0$. The equation gives $m^2 \mid Q - 1$. Absolute values preserve it. The case $V = 0$ is impossible, and $|V| = 1$ gives $|y| = 1$.

For $V > 1, y > 0$, define

$$V' = (2Q-1)V - 2(Q-1)y, \qquad y' = (2Q-1)y - 2QV.$$

Since $m \mid Q - 1$, the new V' is an integer. Expansion preserves the equation. Moreover $y > V$, while

$$Q^2V^2 - (Q-1)^2y^2 = QV^2 + Q - 1 > 0$$

gives $QV > (Q-1)y$. These inequalities imply $-V < V' < V$; the norm excludes $V' = 0$. Finally $2Q - 1 \equiv 1 \pmod{m}$, so y' has exactly the same reduced denominator m. Induction on $|V|$, taking absolute values at each step, reaches $|V| = 1$ without changing m. Thus $m = 1$. □

This is the adjacent-coefficient descent used in the source-69 and incoming root-merger constructions [8]. The coefficients Q and $Q - 1$ are essential; a generic quadratic resultant does not supply this integrality conclusion.

For the sign conditions, define

$$\mathrm{Pos}(p_1, p_2, p_3) = 1 + p_1^2 + p_2^2 + p_3(p_3 + 1).$$

It is positive for every integer triple and represents every positive integer. For the latter claim, apply the classical three-square theorem to $4n - 3$, which is 1 modulo 4 and is not of the excluded form $4^a(8b + 7)$. Its three-square representation has one odd square and two even squares, giving the displayed form after division by four.

Lemma 5.3 (Joint positivity)**.** *For integers d, T, the existence of integers s, p_1, p_2, p_3 with*

$$dT - (s^2 - d - T + 1)^2 = \mathrm{Pos}(p_1, p_2, p_3)$$

is equivalent to $d > 0$ and $T > 0$.

Proof. The equality gives $dT > 0$, so d, T have a common nonzero sign. If $d = -b, T = -e$ with $b, e > 0$, then $(s^2 + b + e + 1)^2 > be$, a contradiction.

Conversely let $d, T > 0$ and put $n = d + T - 1 \geq 1$. Choose an integer square nearest n. Its distance from n has square strictly less than n: between m^2 and $(m+1)^2$, the nearest distance is at most m, and the only nonzero-distance cases have $n > m^2$. Since $n \leq dT$, the residual $dT - (s^2 - n)^2$ is a positive integer. Represent it by Pos. □

In our notation, recovering an integer p_1 with $p_1^2 = \Omega$ restores the joint guard

$$G = dT_\eta - (s^2 - d - T_\eta + 1)^2 - \mathrm{Pos}(p_1, p_2, p_3) = 0. \tag{11}$$

The important feature is that the new equation will recover this guard after the other roots, using premises that do not depend on G.

6 Bounds and residues before using the guard

Lemma 6.1 (Unconditional nondegeneracy)**.** *For every assignment of the thirteen integer inputs in Section 2,*

$$H \neq 0, \qquad \Delta > 0, \qquad L \neq 0, \qquad Q > 1, \qquad \delta \neq 0, \qquad A \geq 1.$$

Furthermore,

$$a \equiv H \equiv \Delta \equiv 0 \pmod 3, \qquad F_2 \equiv A \equiv D_m \equiv 1 \pmod 3, \qquad Q \equiv 0 \pmod 9,$$

and B is a unit congruent to 1 modulo 3 in $\mathbb{Z}_{(3)}$.

Proof. The odd integer $J = 6j + 1$ is nonzero, so $q = \beta J + 1$ is even and $|q| \geq \beta - 1 \geq 16$. The nonzero multiple of four $W = 4(3w + 1)$ has $|W| \geq 4$. Hence

$$|X| \geq 64, \quad |Y| \geq 4096, \quad |a| = |X + 1|\,|Y| \geq 63 \cdot 4096.$$

Thus $\Delta = (a+1)(a+3) > 0$. Because c and $2r + 1$ are odd, $L = (2r + 1)c^2 \neq 0$, and $Q = \Delta^2 L^2 > 1$. The integer $H = 4a + 3$ cannot be zero and is odd. Also Y is even, so a and hence $\Delta - 1$ are even.

Modulo 3, the fixed coefficient and affine restrictions give

$$J \equiv 1, \quad q \equiv 2, \quad W \equiv 1, \quad X \equiv 2, \quad a \equiv 0.$$

The claimed congruences for H, Δ, F_2, D_m, Q follow. Writing $b = XY^2$, we have

$$u = bk, \qquad A = b(b+1)k^2 + 1.$$

For integer b, $b(b+1) \geq 0$. Modulo 3, b is either 0 or 2, so $b(b+1) \equiv 0$. This proves the assertions about A. Finally both $QV^2 - 1$ and $Q - 1$ are -1 modulo 3, so their ratio B is the stated unit. □

The proof neither uses the first or auxiliary Pell equation nor assumes the joint guard. In particular $c = \pm 1$ is permitted at this stage: nonzero oddness is sufficient for the preliminary bounds. Such values will be excluded, if necessary, when the recovered interval guard is combined with the large value of $|Y|$.

7 Soundness of the thirteen-variable equation

Suppose $P_{13} = 0$. By Lemma 6.1, the denominator-clearing factors H, δ are nonzero. The finite norm product therefore supplies algebraic numbers f, τ, y, D, p_1, with signs absorbed into their names, satisfying

$$f^2 = F_2, \quad \tau^2 = A, \quad y^2 = B, \quad D^2 = D_m, \quad p_1^2 = \Omega, \tag{12}$$

and

$$\frac{Z}{2H} = 9f + 3\tau + y + \frac{D}{H} + 27p_1. \tag{13}$$

Repeated roots, reducible quadratics, and $\Omega = 0$ are allowed in this factorization.

The squared orders of the first four terms in (13) are precisely those in (10). The fifth squared order is at least 6, unless $p_1 = 0$. Section 4 makes every root in (12) rational. Lemma 5.1 makes f, τ, D, p_1 integers; Lemma 5.2 makes y an integer. This also shows that any initially negative radicand is impossible at a zero. It is unnecessary, and would be circular, to assume its positivity before root recovery.

Set

$$\gamma = \frac{D - D_0}{H}.$$

Substitute the definition of Z into (13). All terms except the displayed factor $1/2$ are integers, so $2\gamma \in \mathbb{Z}$. Also $H\gamma = D - D_0 \in \mathbb{Z}$. Since H is odd, $\gcd(2, H) = 1$ and $\gamma \in \mathbb{Z}$. The midpoint carrier thus creates no unpaid parity condition.

The integer p_1 restores $G = 0$, and Lemma 5.3 gives $d > 0$ and $\eta(k - \eta) > 0$. The source arithmetic variables can now be restored explicitly.

7.1 Polynomial restoration of the quotients

Put $i = 2r + 1$. The ordinary strong ordinate and scale are

$$I_{\mathrm{old}} = \Delta L, \qquad i_{\mathrm{old}} = \Delta i, \qquad r_{\mathrm{old}} = \frac{i_{\mathrm{old}} - 1}{2}.$$

Both Δ and i are odd, so r_{old} is an integer. Moreover

$$I_{\mathrm{old}} = i_{\mathrm{old}} c^2, \qquad I_{\mathrm{old}}^2 = Q = \Delta(f^2 - 1).$$

The normalized source makes the two quotient restorations especially simple:

$$o = f(cT - R), \qquad j_{\mathrm{aux}} = F_2 T - 1 - \Delta i^2 c^3 R. \tag{14}$$

Direct multiplication gives

$$V + c = fo, \qquad V + R = c j_{\mathrm{aux}}.$$

There is no unproved division by c: the identity $F_2 - 1 = \Delta i^2 c^4$ has already supplied the required quotient as a polynomial.

7.2 Restoration of the remaining source rows

Restore

$$\alpha = q - F - C, \quad \zeta = k - \eta, \quad P = k + u, \quad D_{\mathrm{old}} = D.$$

Then

$$\tau^2 - P^2 + Pk - 1 = \tau^2 - u(u + k) - 1 = 0, \qquad D_{\mathrm{old}}^2 - \Delta c^2 - 1 = 0.$$

The native transport row follows from (2). The index row follows from $k = R+1+hE$. The source's elementary product registers are restored by their defining integer polynomials:

$$q^2 - 1, \quad q^3, \quad c^2, \quad f^2, \quad Q, \quad y^2 - V^2.$$

The auxiliary equations follow from $QV^2 - (Q-1)y^2 = 1$ and (14). For source 70's affine coordinate use $J_{3,\mathrm{old}} = (J-1)/3 = 2j$; the w coordinate is unchanged. Finally represent the recovered positive guard quantities by their private Pos triples.

Every restored coordinate is an integer, and the source's original rows vanish. The signed native soundness theorem therefore supplies a finite ZFC refutation. This establishes the forward existential implication without assuming either consistency or the absence of a refutation.

8 Canonical completeness and the normalized strong root

Begin with a finite ZFC refutation and take the canonical family (9). Its native $J = (q-1)/\beta$ is a finite repunit in the even base $2^{1064921}$, hence is odd. Residue orientation gives $J \equiv 1 \pmod 3$, so $J = 6j + 1$ for an integer j. The main Pell parameter $a + 2$ is even. The Pell recurrence modulo 2 then shows that $c = \psi_{a+2}(R)$ is odd because R is odd. Thus $c = 2c_0 + 1$ is a permissible syntactic restriction on this family.

We must prove that the stronger normalization $L = (2r+1)c^2$ does not destroy completeness.

Lemma 8.1 (Odd normalized strong choice)**.** *Under (9), there are positive integers f, L with*

$$f^2 - \Delta L^2 = 1, \qquad L/c^2 \text{ an odd integer.}$$

Proof. Let $b = a + 2$, choose $m = cR$, and set $f = \chi_b(m)$, $L = \psi_b(m)$. The Pell norm gives the first equation. Composition gives

$$L = \psi_b(R)\psi_{\chi_b(R)}(c) = c\psi_D(c), \qquad D = \chi_b(R).$$

Since $D^2 - \Delta c^2 = 1$, we have $D^2 \equiv 1 \pmod c$. The recurrence consequently gives

$$\psi_D(n) \equiv nD^{n-1} \pmod c.$$

At $n = c$ this proves $c^2 \mid L$. Both c and R are odd, so m is odd. As b is even, the recurrence modulo 2 makes L odd. Its integer quotient by the odd square c^2 is therefore odd. □

Choose r with $L/c^2 = 2r + 1$. The remaining issue is the stronger form of the auxiliary quotient, which contains f^2 rather than f.

Lemma 8.2 (Fresh auxiliary witnesses)**.** *Let $R \geq 3$, $R \equiv 3 \pmod 4$, $c = \psi_b(R)$, and suppose $f, L > 0$ satisfy $f^2 - (b^2-1)L^2 = 1$ with $c \mid L$. Put $S = (b^2-1)L$ and $Q = S^2$. There are positive integers T, y such that*

$$V = f^2(cT - R) - c, \qquad QV^2 - (Q-1)y^2 = 1.$$

Proof. This is source 61's positive auxiliary reconstruction [3]. Its explicit choice is

$$V_* = \frac{\chi_S(R)}{S}, \qquad y_* = \psi_S(R).$$

The odd-index Pell identities make V_* an integer and give

$$V_* \equiv -c \pmod{f^2}, \qquad V_* \equiv -R \pmod{c}.$$

Here $S^2 \equiv 1 - b^2 \pmod{f^2}$, and $R \equiv 3 \pmod 4$ fixes the required signs. The strong equation gives $f^2 \equiv 1 \pmod c$, hence $\gcd(f^2, c) = 1$. Therefore

$$T = \frac{V_* + c + Rf^2}{cf^2}$$

is an integer and has $f^2(cT - R) - c = V_*$. The source identities give the auxiliary norm with $y = y_*$; the chosen positive branches make T, y positive. □

The two lemmas choose fresh strong and auxiliary witnesses. They do not claim that every earlier source solution had the stronger quotient divisibility. The new choices leave q, F, C, k, η unchanged. Thus the canonical determinant $d > 0$ and interval product $T_\eta > 0$ remain valid. Lemma 5.3 supplies s, p_1, p_2, p_3.

Let τ, γ be the canonical first root and main congruence quotient. Define the integer

$$x = 18f + 6\tau + 2y + 2\gamma + 54p_1 - 18(a+2)L - 3(2u+k) - 2V. \tag{15}$$

Substitution makes one factor in (8) vanish. Therefore a finite refutation supplies an integer zero of P_{13}. Together with Section 7, this proves Theorem 1.1.

9 Three companion constructions

9.1 Fourteen integers

Restore p_1 as an independent coordinate, retain c itself as an independent integer, and omit the syntactic substitution $c = 2c_0 + 1$. Use the same normalized strong expressions and the guard G in (11). Define the raw four-root polynomial

$$N_{14} = \delta^8 \mathcal{N}_4(Z; 324H^2F_2, 36H^2A, 4H^2B, 4D_m).$$

Then set

$$P_{14} = N_{14} + G^2 + N_{14}G^2.$$

The equation is equivalent to $N_{14} = G = 0$, since $(1 + N_{14})(1 + G^2) = 1$ and the second factor is a positive integer. Here the guard is available before root recovery. It gives

$$|k| \geq 2, \quad 0 < |\eta| < |k|, \quad |c| = |Yk + \eta| > (|Y| - 1)|k| > 1.$$

Thus $L \neq 0, Q > 1$, and the preceding four-root argument applies. Its exact degree is 696, and its compact circuit has 151 operations.

9.2 Sixteen integers and the fixed multiplier three

For the remaining companions retain an independent integer f and put

$$J = 3j + 1, \quad L = (2r+1)c^2, \quad Q = L^2, \quad F_2 = f^2, \qquad V = fcT - f^2R - c.$$

Keep the separate strong row $S_0 = Q - \Delta(f^2 - 1)$. All other common abbreviations have the same meanings. The seventeen-integer CRT parent supplies this interface [9].

The guard gives $|c| > 1, Q > 1$. Since $\Delta \equiv 0 \pmod 3$, the strong row makes $3 \mid L$ and $9 \mid Q$. Thus the fixed multiplier 3 separates the first and auxiliary roots. Put

$$\begin{aligned} t_0 &= 2u + k, & \ell_0 &= k^2 - 4, \\ J_*(x) &= \delta x^2 + \delta(6t_0 + 4V)x + 9\delta\ell_0 - 4(V^2 - 1), \\ H_* &= 12\delta t_0 V, \\ N_{16}(x) &= J_*(x)^2 + 2H_* J_*(x) + 576\delta A(1 - V^2) + 144\delta^2 \ell_0 V^2. \end{aligned}$$

An exact identity gives

$$N_{16}(x) = \delta^2 \mathcal{N}_2(x + 3t_0 + 2V; 36A, 4B).$$

This replaces τ, y by $x = 6\tau + 2y - 3t_0 - 2V$. Root separation, monicity, and Lemma 5.2 prove the converse. The sum of squares of N_{16}, S_0, G, and $(D_0 + H\gamma)^2 - D_m$ is P_{16}, of degree 108 in sixteen integers and 132 operations.

9.3 Fifteen integers

Eliminate γ as well, using

$$N_{15}(x) = H^8 \prod_{(H\gamma + D_0)^2 = D_m} N_{16}(x - 2\gamma).$$

The product includes both roots and denotes its denominator-free polynomial expansion. The added weighted root D/H has order $-v_3(H)$, below the first two. The same rational extraction and odd-H midpoint argument recover an integer γ. With $S = S_0^2 + G^2$, set

$$P_{15} = N_{15} + S + N_{15}S.$$

This equation is equivalent to the three required rows. Its exact degree is 244, and its circuit has 137 operations. These two companions retain the existing CRT parent's rank and divisibility proof; those dependencies are not replaced by finite samples.

10 Division-free evaluation and arithmetic counts

A norm product is a compact algebraic specification, but a concrete straight-line evaluator makes the equation independently reproducible. Here is the four-root evaluator used inside the thirteen-variable circuit.

For inputs $Z, A_1, A_2, A_3, C_*, \delta$, set

$$\begin{aligned} \sigma &= A_1 + A_2 + A_3, & \pi &= A_1A_2 + (A_1 + A_2)A_3, & \rho &= A_1A_2A_3, \\ a_* &= \delta(Z^2 - \sigma) + C_*, & t_* &= \delta C_* Z^2, \\ k_* &= a_*^2 + 4(t_* - \delta^2\pi), & r_* &= 16\delta^3\rho, \\ b_* &= a_* k_* - r_*, \\ e_* &= k_*^2 + 16 t_* a_*^2 - 4r_*(\delta Z^2 + C_*), \\ \mathcal{Q} &= e_*^2 - 64 t_* b_*^2. \end{aligned}$$

Direct multiplication of formal norm factors proves

$$\mathcal{Q} = \delta^8 \mathcal{N}_4(Z; A_1, A_2, A_3, C_*/\delta).$$

Use

$$A_1 = 324H^2F_2, \quad A_2 = 36H^2A, \quad A_3 = 4D_m, \quad C_* = 4H^2(QV^2 - 1).$$

Evaluate $\mathcal{Q}$ at $Z - \omega$ in the quadratic algebra $\omega^2 = 2916H^2\Omega$. If its result is $E_* + O_*\omega$, then

$$P_{13} = E_*^2 - 2916H^2\Omega\, O_*^2.$$

For this calculation addition is componentwise and multiplication is

$$(e, o)(e', o') = (ee' + 2916H^2\Omega\, oo',\ eo' + oe').$$

The pair entries are derived polynomial expressions; they do not add unknowns. This gives an entirely division-free definition of (8).

The implementation shares repeated nodes, uses fixed powers only, and expands every power into multiplication. The compact circuits have the following verified operation splits:

	Additions	Subtractions	Multiplications	Total
P_{13}	47	28	130	205
P_{14}	38	23	90	151
P_{15}	33	24	80	137
P_{16}	33	24	75	132

These are exact counts of supplied evaluators, not proofs of minimum circuit size or estimates of bit complexity. Operations on million-bit fixed integers are not unit-cost in an implementation concerned with running time. A separate coefficient-based implementation expands the successive quadratic norms in their carrier coordinate and provides an independent route for verification and degree certification.

11 Exact degree proofs

Naive degree propagation through a compact circuit can overestimate the degree because it misses cancellations. The compact fourteen-variable circuit, for example, has a propagated bound of 792, while its actual degree is 696. We use the independent coefficient construction to prove upper bounds, and exact finite-field specializations to certify nonzero leading coefficients after fixing the actual numerical ports.

11.1 Sixteen and fifteen variables

The coefficient construction gives upper bounds 54 for the norm N_{16} and 156 for N_{15}. The former is the largest residual degree in its sum of squares, giving degree at most 108. In the fifteen-variable finalizer, G has degree 44, so $S = S_0^2 + G^2$ has degree 88, and $N_{15} + S + N_{15}S$ has degree at most 244. The matching nonzero certificates below make these bounds exact.

11.2 The centered strong root and degree 696

Immediately before the final strong-root norm in the fourteen-variable coefficient construction, write the degree-eight carrier polynomial as $\sum_{i=0}^{8} a_i x^i$. Direct arithmetic propagation bounds the coefficient degrees by

$$(292, 295, 294, 277, 260, 237, 214, 191, 168).$$

The final root is centered at $f_0 = (a+2)L$. Writing $w_f = f - f_0$, its exact equation is

$$w_f^2 + 2f_0 w_f + (L^2 - 1) = 0, \qquad \deg f_0 \le 9, \quad \deg(L^2 - 1) \le 6. \tag{16}$$

The identity $f_0^2 - F_2 = L^2 - 1$ is responsible for the degree saving.

Scale every independent coordinate along a fixed real direction by a parameter $t \to +\infty$. On a nonempty open set of directions, the degree-nine leading coefficient of f_0 is nonzero. The roots of (16) are eventually real: f_0^2 has degree eighteen while $L^2 - 1$ has degree six. One root is $O(t^9)$. By the product of the roots, or the conjugate quotient formula, the other is $O(t^{-3})$. The two arguments $x - 18w_f$ are therefore $O(t^9)$ and $O(t)$. Their polynomial values have product of order at most

$$t^{M_9+M_1}, \qquad M_9 = \max_i(\deg a_i + 9i) = 312, \quad M_1 = \max_i(\deg a_i + i) = 296.$$

Thus the raw norm has degree at most 608. This is a global degree bound: a homogeneous polynomial component of larger degree would vanish on the open set of directions just used, and hence would be identically zero. The argument does not merely estimate the polynomial at one chosen numerical ray.

The guard has exact degree 44, since its leading term is the negative square of the degree-22 product $\eta(k - \eta)$. Therefore

$$\deg P_{14} \le 608 + 2 \cdot 44 = 696.$$

11.3 The guard norm and degree 1,392

Before folding the guard root, the degree-sixteen raw norm as a polynomial in its carrier has the following coefficient-degree bounds:

$$(612, 607, 606, 597, 588, 571, 554, 537, 520,$$
$$497, 474, 451, 428, 405, 382, 359, 336).$$

The affine substitution $c = 2c_0 + 1$ does not increase them. Since $\deg \Omega \le 44$, assign weight 22 to a formal root $\omega^2 = \Omega$. Either substituted factor has weighted degree at most

$$\max_{0 \le i \le 16} (\deg a_i + 22i) = 696.$$

Its quadratic norm therefore has degree at most $1,392$. This can also be read as direct degree propagation in the quadratic algebra; it does not require that the formal root already be real or integral.

11.4 Nonzero leading coefficients on the actual coefficient slice

The checker replaces each integer coordinate by a recorded integer slope times t, reduces all six actual coefficient payloads modulo the prime 65521, and performs exact polynomial convolution. A nonzero coefficient at the upper-bound degree certifies that the corresponding homogeneous component is not zero.

The prime is checked by trial division. The convolution uses arbitrary precision integers before modular reduction. The exact slope vectors, coefficient residues, intermediate coefficient bounds, and outputs are included. These are rigorous finite lower certificates, not probabilistic claims of polynomial identity. Together with the symbolic upper bounds they establish all exact degrees in Table 1.

	Norm degree	Norm coefficient	Final degree	Final coefficient
16 integers	54	17,367	108	19,526
15 integers	156	29,172	244	16,359
14 integers	608	32,234	696	50,632
13 integers	1,392	56,423	1,392	56,423

Table 3: Nonzero leading coefficients modulo 65521. Each row uses its own fully recorded slope specialization.

12 Verification and reproducibility

The artifact has two deliberately different implementations of the norm construction. The compact one uses the closed four-root formula and quadratic pair arithmetic. The coefficient implementation treats the carrier as a polynomial variable and takes successive quadratic norms. The identities defining both are proved algebraically. Independent formal sign-product expansions check the universal norm formulas, while modular comparisons with arbitrary coordinate values check transcription of the literal circuits.

The additional checker `independent_audit/check_primary_formula.py` imports neither circuit builder. It independently states all twenty-eight source abbreviations and the closed norm evaluator, then canonicalizes the actual 205-gate packet by elementary ring identities. This proves exact equality with the displayed P_{13}, without sampling. In particular it checks the passage from δ^8 to δ^{16}, all five radicand scales, the guard fold, and agreement of the actual source-34 coefficient bytes.

The finite checks establish coefficient integrity, declared variable counts, absence of undeclared inputs, node liveness, operation counts, exact algebraic identities, and degree certificates. Normal and optimized Python runs retain the active checks. They do not search for a solution of the consistency equation or prove its unsolvability.

The archive includes the complete LaTeX article, its compiled PDF, the two circuit implementations, literal polynomial packets, coefficient payloads and rebuilding data, provenance and license notices, verification scripts, and receipts. Its source snapshot is fixed even if the repository subsequently advances. The ordinary proof in this article is also retained in a readable proof file so that the algebraic construction can be reviewed independently of the PDF layout.

A rebuild should run the included verification driver before rebuilding the report. From the extracted root, use `python3 -O verify_all.py`; its checks remain active under Python optimization. The mathematical checks use ordinary Python; the report uses a standard `pdflatex` installation and two passes. PDF byte identity is not required, since creation metadata and TeX versions can vary. Polynomial packet identity and coefficient hashes are the stable reproducibility targets.

13 Ancillary exact improvements to ten-integer evaluators

A separate part of the work applies three exact identities to the current ten-integer shear circuits [10, 5]. These changes preserve each polynomial identically, so its witness domain, consistency theorem, and exact degree are inherited without a new semantic reduction.

The first reuses already available powers:

$$S^5 = (S^2)(S^3).$$

The second changes the placement of a two-root norm:

$$\begin{aligned}(Z^2 - \Sigma^2(a_0 + a_1))^2 - 4\Sigma^4 a_0 a_1 \\ = (Z^2 + \Sigma^2(a_0 - a_1))^2 - 4Z^2\Sigma^2 a_0.\end{aligned}$$

The third factors a common block; writing c_* for the existing capacity register and $D = C - KL$, it is

$$\begin{aligned}8c_*^3 gK^2 - g^2(32D^2 c_*^3 + g^2 K^2) \\ = 8gc_*^3(K^2 - 4gD^2) - g^4 K^2.\end{aligned}$$

Each identity saves one operation in the supplied shared evaluator, for three operations per circuit.

Variant	Exact degree	Prior operations	New operations
Predecessor	1,245,324	480	477
Linear shear	1,179,788	484	481
Cubic shear	1,114,252	487	484
Six-term shear	1,081,125	521	518

The optimization packets retain source hashes and the exact substituted circuits. Modular checks are used only as transcription diagnostics; the three displayed ring identities are the proofs. These small improvements are separate from the thirteen-variable prime-separation construction and should not be described as the same type of advance.

14 Further research questions

How far can local-order separation be generalized? The proof uses the prime 3 because the source's existing affine restrictions make a, H, Δ divisible by 3, while the relevant radicands are units. Other encodings may expose a different prime or several primes. A useful general result would minimize fixed root weights subject to a proved noncancellation condition, while accounting for the operation cost of those weights and the cost of new congruence restrictions.

Can a second guard coordinate be removed? The fifth root works because one guard coordinate appears as a separate square and the other arithmetic roots remain recoverable without the guard. Repeating that argument is not automatic: the remaining guard coordinates share a single quadratic representation, so they do not supply independent radicands. A further reduction needs a new complete parameterization or another arithmetic invariant, rather than simply adding more formal norm factors.

Can normalized strong coordinates help other degree slices? The choice $L = (2r + 1)c^2$ makes quotient restoration polynomial and removes a preliminary rank dependency from the new soundness proof. It also changes the degrees of F_2, Q, V. Retaining one intermediate coordinate might lower the degree or operation count at fifteen or sixteen variables. Such variants should be compared as complete literal circuits, including every restored divisibility or sign condition.

Can degree certificates be extracted automatically from norms? The centered strong-root argument is a small example of using the different growth rates of conjugate roots to prove cancellation-aware degree bounds. An exact tool that combines coefficient bounds, Newton polygons, and a finite nonzero specialization could make such bounds routine. Its output should distinguish a symbolic upper-bound proof from a numerical lower certificate, as the present construction does.

What is the appropriate notion of simplicity? Variable count, degree, circuit operations, expanded monomial count, coefficient height, and total encoded description length are different objectives. The current native coefficient inputs occupy 615,883 raw bytes, although they compress well. A comparison in a fixed description language should count the coefficient recipe or the literal bytes consistently. It should not infer bit complexity from the number of arithmetic gates.

Can the computational source be replaced? The user explicitly allowed non-K215 and non-Jones routes. The present advance is in the arithmetic representation of a calibrated source. A direct proof-language or equiconsistent-theory encoding would need a complete source predicate, a proved bridge to the same syntactic consistency sentence, and all of its sequence and arithmetic encodings materialized before a fair size comparison. The prime-separation lemma could be useful in such a compiler without forcing it to reuse this native tableau.

What should be formalized next? The elementary sign-product lemma, rational extraction, adjacent Pell descent, and midpoint integrality are focused independent targets. The literal norm identities and quotient restorations are suitable for a polynomial reflection checker. A complete formal result would then need to connect those statements to the actual coefficient bytes and to the inherited native and syntactic-calibration theorems. Closing that boundary is a separate deliverable from the finite circuit audits reported here.

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[9] ProveIt incoming research. *Pell congruences for Diophantine consistency*, archive `con_zfc_pell_congruences_2026-10-23`. The CRT quotient proof and `transport/degree88_17.json` provide the seventeen-integer parent for the companions. Inspected under `docs/incoming` at the snapshot in [1].

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